Model parameters

Every number the model needs, with its source. Values are mean ± SD where the label gives one.

Norethindrone (minipill, 0.35 mg daily, no break)

Parameter	Value	Source
Half-life (single dose)	7.7 ± 0.5 h	Norethindrone label
Time to peak (Tmax)	1.2 ± 0.5 h	Norethindrone label
Peak level (Cmax, single dose)	4,817 ± 1,533 pg/mL	Norethindrone label
Average level over 24 h	885 ± 250 pg/mL	Norethindrone label
Level at 24 h	130 ± 47 pg/mL	Norethindrone label
Volume of distribution	about 4 L/kg	Heather label
Clearance	about 600 L/day	Heather label
Protein binding	36% SHBG, 61% albumin	Norethindrone label

Levonorgestrel (combined pill progestin)

Parameter	Value	Source
Dose	0.10 mg (Alesse) or 0.15 mg (Seasonale) daily	Labels below
Half-life, single dose	about 30 h	Seasonale label
Half-life, steady state	36 ± 13 h	Alesse label
Time to peak at steady state	1.5 ± 0.5 h	Alesse label
Steady-state peak (0.10 mg)	6.0 ± 2.7 ng/mL	Alesse label
Steady-state low (0.10 mg)	1.9 ± 1.0 ng/mL	Alesse label
Steady state reached	from day 19	Alesse label
Rise over the cycle	3- to 4-fold in second half	Levette label
Bound to SHBG	42–68%	Levette label
SHBG increase on the pill	about 1.7-fold	Levette label

Ethinyl estradiol (combined pill estrogen)

Parameter	Value	Source
Dose	0.02 mg (Alesse) or 0.03 mg (Seasonale) daily	Labels below
Half-life, single dose	about 15 h	Seasonale label
Half-life, steady state	18 ± 4.7 h	Alesse label
Oral bioavailability	about 43%	Seasonale label
Single-dose peak (0.02 mg)	62 ± 21 pg/mL at 1.5 ± 0.5 h	Alesse label
Steady-state peak (0.02 mg)	77 ± 30 pg/mL	Alesse label
Steady-state low (0.02 mg)	10.5 ± 5.1 pg/mL	Alesse label
Steady state reached	from day 6	Alesse label
Binding and metabolism	95–97% albumin, not SHBG; induces SHBG; broken down partly by CYP3A4	Seasonale label

Timing rules to compare against

Pill	Counts as missed	What to do	Source
Minipill (norethindrone)	More than 3 h late	Take it ASAP; backup until 2 days of on-time pills	CDC summary (UCSF)
Combined pill, one pill	Under 48 h since it was due	Take it ASAP; no backup needed	CDC summary (UCSF)
Combined pill, two or more	48 h or more	Take the latest pill; backup for 7 days	CDC summary (UCSF)

Numbers I'll have to estimate

- **Absorption rate.** No label gives it directly, so I'll pick one that reproduces each drug's time to peak (about 1 to 1.5 h).
- **Volume for levonorgestrel and ethinyl estradiol.** I can scale the model so single-dose peaks match the labels instead.
- **How fast SHBG rises and how much it slows levonorgestrel.** I'll tune these so SHBG roughly reaches 1.7-fold and levonorgestrel rises 3- to 4-fold, and say clearly that they're calibrated, not measured.